

CYBERSECURITY NETWORK DESIGN FOR KAFITECH SOLUTIONS USING PACKET TRACER

1. EXECUTIVE SUMMARY

Kafitech Solutions required a cyber-resilient network architecture to securely connect its five departments (Software Development, HR, Finance, IT Support, and Management) while accommodating guest access. This project implemented a VLAN-segmented, ACL-controlled network in Cisco Packet Tracer, enforcing strict security policies at Layer 2 and Layer 3 to prevent unauthorized access, data leaks, and network attacks.

Key Design Features

a. Logical Segmentation & Access Control

- 1 Departmental VLANs: Each department isolated in its own subnet (VLANs 10-50) with dedicated IP ranges.
- 1 Guest Isolation: Guest traffic restricted to VLAN 70 (192.168.70.0/24), with internet-only access.
- 1 ACL Enforcement:

HR ↔ Finance mutual blocking

Developers limited to their VLAN + DMZ web server (203.0.113.0/24)

IT Support denied access to Management VLAN

Guests blocked from internal networks

b. Secure Wireless Connectivity

- 1 Corporate WiFi (5GHz):

Hidden SSID, WPA2-PSK, MAC filtering

Connected to internal VLANs (VLAN 60)

- 1 Guest WiFi (2.4GHz):

WPA2-PSK, restricted to VLAN 70 (internet only)

c. Layer 2 Security (Switch Hardening)

- 1 Port Security: Sticky MAC (1 device per port)
- 1 BPDU Guard: Prevents rogue switch attacks
- 1 DHCP Snooping: Blocks rogue DHCP servers
- 1 Dynamic ARP Inspection (DAI): Prevents ARP spoofing
- 1 Storm Control: Mitigates broadcast storms

d. Device Hardening & Monitoring

- 1 SSH-Only Access: Telnet disabled
- 1 Encrypted Passwords: Enable secret hashing
- 1 Banner Messages: Legal warnings for unauthorized access
- 1 Syslog & SNMP: Centralized logging and monitoring

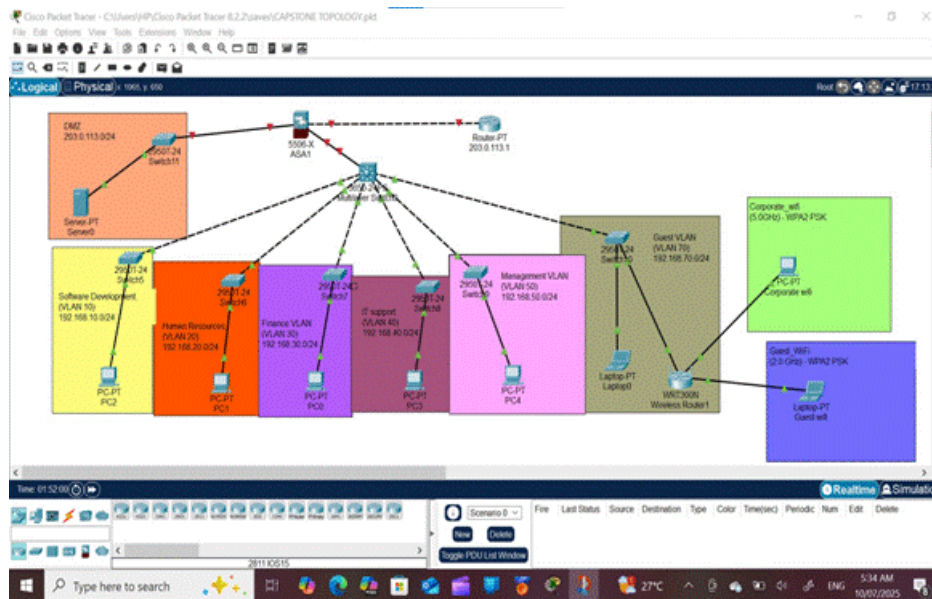
e. DMZ & Internet Access

- 1 DMZ Web Server (203.0.113.10): Accessible only to Developers
- 1 Guest Internet Access: Restricted via ACLs (no internal access)

Validation & Compliance

- 1 Ping Tests: Confirmed permitted/blocked traffic flows (included in report).
- 1 Security Audits: Simulated attacks (rogue DHCP, ARP spoofing) were mitigated by Layer 2 protections.
- 1 Policy Enforcement: Password policies, USB restrictions, and auto-updates documented for real-world deployment

2. NETWORK TOPOLOGY DIAGRAM



3. IP ADDRESSING AND LAN TABLE

Department	VLAN ID	Subnet	Gateway
Software Development	10	192.168.10.0/24	192.168.10.1
Human Resources	20	192.168.20.0/24	192.168.20.1
Finance	30	192.168.30.0/24	192.168.30.1
IT Support	40	192.168.40.0/24	192.168.40.1
Management	50	192.168.50.0/24	192.168.50.1
Guest Wireless	70	192.168.70.0/24	192.168.70.1
DMZ (Web Server)	N/A	203.0.113.0/24	203.0.113.1

4. ACL LIST WITH PURPOSE

Based on project requirements, here are the named ACL configurations for inter-VLAN traffic control. All ACLs are applied inbound on router subinterfaces.

ACL Name	Source Network	Destination Network	Protocol	Action	Purpose
HR_ACL	192.168.20.0/24	192.168.30.0/24	IP	DENY	Block HR → Finance access
	192.168.20.0/24	any	IP	PERMIT	Allow HR to all other networks
FINANCE_ACL	192.168.30.0/24	192.168.20.0/24	IP	DENY	Block Finance → HR access
	192.168.30.0/24	any	IP	PERMIT	Allow Finance to all other networks
DEV_ACL	192.168.10.0/24	203.0.113.0/24	IP	PERMIT	Allow Developers → DMZ Web Server
	192.168.10.0/24	192.168.0.0/16	IP	DENY	Block Developers from other internal VLANs (implicit deny all others)
IT_SUPPORT_ACL	192.168.40.0/24	192.168.50.0/24	IP	DENY	Block IT Support → Management
	192.168.40.0/24	any	IP	PERMIT	Allow IT Support to all other networks
GUEST_ACL	192.168.70.0/24	192.168.70.1/32	IP	PERMIT	Allow Guests → their gateway (for internet access)

ACL Name	Source Network	Destination Network	Protocol	Action	Purpose
	192.168.70.0/24	192.168.0.0/16	IP	DENY	Block Guests from internal VLANs
	192.168.70.0/24	any	IP	PERMIT	Allow Guests → Internet (via DMZ/NAT)
MGMT_ACL	any	any	IP	PERMIT	Allow Management unrestricted access (no restrictions)

5. LAYER 2 SECURITY CONFIGURATION SUMMARY

Feature	Configuration	Protected Against
Port Security	Sticky MAC (limit 1), violation shutdown	Unauthorized device connections
BPDU Guard	Enabled globally on edge ports	Rogue STP topology changes
DHCP Snooping	Trusted uplinks, VLAN-specific	Rogue DHCP servers
DAI	ARP validation + trusted ports	ARP spoofing
Storm Control	50% broadcast threshold	Broadcast storms
Unused Ports	Shutdown + assigned to VLAN 999	Physical tampering

6. WIFI SECURITY IMPLEMENTATION

SSID	Frequency	VLAN	Purpose	Security Protocol	Additional Protections
Corporate_WiFi	5.0 GHz	VLAN 60 (Internal)	Employees (Dev, HR, Finance, IT, Mgmt)	WPA2-PSK (AES-CCMP)	MAC Filtering, Hidden SSID
Guest_WiFi	2.4 GHz	VLAN 70 (Isolated)	Visitors/Contractors	WPA2-PSK (AES-CCMP)	Client Isolation, No Intra-VLAN

7. SAMPLE CONFIGURATION SNIPPETS

Configuration Snippet 1:

```
Shell
Switch> enable

Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name Software_Dev
Switch(config-vlan)# exit
Switch(config)# interface range fa0/1-24
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit
Switch(config)# interface gig0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan
10,20,30,40,50,70
```

Configuration Snippet 2:

Shell

```
Switch> enable
Switch# config t
Switch(config)# vlan 30
Switch(config-vlan)# name Finance_vlan
Switch(config-vlan)# exit
Switch(config)# interface range fa0/1-24
Switch(config-if-range)# switchport access vlan 30
Switch(config-if-range)# exit
Switch(config)# interface gig0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 30,40,50,70
```

Configuration Snippet 3:

Shell

```
Switch> enable
Switch# config t
Switch(config)# vlan 70
Switch(config-vlan)# name Guest_Vlan
Switch(config-vlan)# exit
Switch(config)# interface range fa0/1-24
Switch(config-if-range)# switchport access vlan 70
Switch(config-if-range)# exit
Switch(config)# interface gig0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 70
```

8. SYSLOG CONFIGURATION (ROUTER/SWITCHES)

Bash:

! Enable logging to syslog server

logging host 192.168.40.10

logging trap debugging service timestamps log datetime

Logged Events:

Interface status changes (up/down).

ACL denials.

Port security violations.

SNMPv3 (Simulated)

Bash:

! Router/Switch SNMP setup

snmp-server group AdminGroup v3 priv

snmp-server user Admin AdminGroup v3 auth sha AuthPass123 priv aes 256 PrivPass456

snmp-server host 192.168.40.10 version 3 priv Admin

Monitored Metrics:

Bandwidth usage per VLAN.

Device uptime.

ACL Logging

Bash:

ip access-list extended HR_ACL

deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255 log ! Log HR→Finance denials

9. CHALLENGES AND MITIGATIONS

a. Packet Tracer Limitations

Challenge	Mitigation
No Syslog Server Emulation	Use a generic server with logging service enabled; document real-world SIEM (Splunk) setup in the report.
SNMP Polling Not Functional	Simulate SNMP with CLI show commands; recommend SNMPv3 + SolarWinds in production.
No WPA3/802.1X Support	Use WPA2-PSK + MAC filtering; document WPA3-Enterprise as a future upgrade.
Limited ACL Logging	Use log keyword in ACLs and manually track hits via show access-lists.

b. Security Design Challenges

Challenge	Mitigation
MAC Filtering Scalability	Manual MAC entry is tedious for large networks; recommend RADIUS (802.1X) for enterprises.
Guest WiFi Isolation	Use VLAN 70 + ACLs to block internal access; enable client isolation on SSID.
HR/Finance VLAN Segregation	Strict ACLs (deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255).
Single Point of Failure (Core Router)	Recommend HA (HSRP) in real deployments.

c. Operational Challenges

Challenge	Mitigation
Password Management	Use enable secret + SHA-256 hashing; document password rotation policies.
Port Security False Positives	Configure errdisable recovery to auto-reenable ports after 300s.
Complex ACL Maintenance	Document rules in a table (see ACL section); use named ACLs for readability.
No Endpoint Compliance Checks	In real networks, enforce via MDM (Intune/Jamf) or NAC (Cisco ISE).

d. Performance & Scalability

Challenge	Mitigation
Broadcast Storms	Enable storm control (storm-control broadcast level 50).
Inter-VLAN Latency	Ensure router-on-a-stick has sufficient bandwidth (1Gbps uplinks).
Wireless Congestion	Separate 5GHz (Corporate) and 2.4GHz (Guest); recommend additional APs.

10. RECOMMENDATION FOR REAL DEPLOYMENT

a. Network Security Enhancements

Recommendation	Purpose	Tools/Technologies
Replace ACLs with Firewall	Stateful inspection, L7 filtering, and threat prevention.	Cisco ASA, FortiGate, Palo Alto NGFW
Upgrade to WPA3-Enterprise	Stronger encryption and individual user authentication.	Cisco ISE, FreeRADIUS
Implement 802.1X (NAC)	Device authentication for both wired/wireless networks.	Cisco ISE, Aruba ClearPass
Deploy IPsec for Inter-VLAN	Encrypt sensitive traffic (e.g., HR→Finance).	Cisco Crypto Maps, OpenVPN

b. Monitoring & Compliance

Recommendation	Purpose	Tools/Technologies
SIEM Integration	Centralized logging, anomaly detection, and compliance reporting.	Splunk, IBM QRadar, ELK Stack
NetFlow/sFlow Analysis	Real-time traffic visibility and DDoS detection.	Cisco Stealthwatch, PRTG
Endpoint Compliance Checks	Enforce security policies (patches, antivirus) on devices.	Microsoft Intune, Jamf, Cisco AMP

c. Redundancy & High Availability

Recommendation	Purpose	Implementation
HSRP/VRRP for Core Router	Failover during router outages.	Cisco HSRP, VRRP
StackWise/VSS for Switches	Eliminate single points of failure in access layer.	Cisco StackWise, Virtual Switching System
Dual ISP Connections	Backup internet connectivity.	BGP, SD-WAN

d. Wireless Network Improvements

Recommendation	Purpose	Implementation
Captive Portal for Guests	Acceptable Use Policy (AUP) and bandwidth throttling.	PacketFence, Aruba Captive Portal
Wi-Fi 6 (802.11ax) Upgrade	Higher throughput and better device density.	Cisco Catalyst 9100, Aruba AP-500 Series
RF Heat Mapping	Optimize AP placement for coverage/performance.	Ekahau, NetSpot

e. Endpoint & Data Protection

Recommendation	Purpose	Tools/Technologies
DLP (Data Loss Prevention)	Block unauthorized data transfers (e.g., USB, cloud).	Symantec DLP, Microsoft Purview
Disk Encryption	Protect data at rest on laptops/desktops.	BitLocker, FileVault
Zero Trust Architecture	Micro-segmentation and least-privilege access.	Zscaler, Duo Security

f. Physical Security

Recommendation	Purpose	Implementation
Secure Server Room Access	Prevent unauthorized physical access to network devices.	Biometric locks, CCTV
UPS & Environmental Controls	Protect against power surges and overheating.	APC Smart-UPS, Liebert Cooling

11. PING SCREEN SHOTS

